Topology

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Intro

The following are my notes for Topology which I will (hopefully) be taking in the fall of 2026. As these are my own personal notes I may skip ceirtain topics I already understand, be sloppy at times and this may cause trouble for anyone who wishes to use these notes for themselves so keep that in mind if you want to use these.

1 Topological spaces and continuity

We recall from Real Analysis that for a metric space (X, d) we have a notion of distance and can start describing open sets $U \subseteq X$ as sets where for every $x \in U$ there exists $\varepsilon > 0$ such that

$$B(x; \varepsilon) \subseteq U,$$

meaning that for any x in U there is some open ball around it contained entirely in U.

In Hilbert spaces, normed spaces, $\mathbb{R}^n$ (different types and instances of metric spaces), etc., this is the usual idea: an open set is one where every point has some wiggle room around it.

Topology starts from the observation that many arguments in Analysis don't really use the distance $d(x, y)$. They only use collections of open sets.

So instead of letting open sets be generated by some metric, why don't just specify which sets are open?

Intuitively, a topology on a set X is a choice about which sets are "open", whatever that may mean. Informally a topology tells us which points are "near" eachother without really specifing quantitatively what that may mean.

So we want the least structure possible to be able to talk about things like continuity, connectedness, compactness, convergence, etc.

Definition 1.1 (Topological space). *Let X be a set. A **topology** on X is a collection $\tau \subseteq \mathcal{P}(X)$ satisfying:*

1. *$\emptyset, X \in \tau$.*
2. *If $\{U_i\}_{i \in I}$ is any collection of elements in τ, then*
$$\bigcup_{i \in I} U_i \in \tau.$$
3. *If $U_1, \dots, U_n \in \tau$, then*
$$U_1 \cap \dots \cap U_n \in \tau.$$

*The pair (X, τ) is called a **topological space**. The elements of τ are called **open sets**.*

We will often refer to a topological space by just the set if it is clear what the topology on X is.

The axioms are not arbitrary, but abstracted from metric spaces. In metric spaces we already have these properties even if they are not explicitly required. For instance in $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty}\left(-\frac{1}{n},\frac{1}{n}\right)=\{0\}.$$

Each interval is open, but $\{0\}$ is not.

Intuitively, finite intersections preserve some positive amount of wiggle room while infinite intersections can squeeze it all away.

Examples of topological spaces

Example 1.1 ($\mathbb{R}$). *Let $X=\mathbb{R}$. The usual topology is the collection of subsets $U\subseteq\mathbb{R}$ such that for every $x\in U$, there exists $\varepsilon>0$ with*

$$(x-\varepsilon,x+\varepsilon)\subseteq U.$$

This is the topology we know from Real analysis. Examples of open sets include:

$$(0,1),\;\;(-3,7),\;\;\mathbb{R},\;\;\emptyset,$$

and also things like

$$(0,1)\cup(2,5)\cup(10,\infty).$$

Non-examples:

$$[0,1],\;\;\mathbb{Q},\;\;\{0\}.$$

Note that $\mathbb{Q}$ can be made into a topological space by giving it the subspace topology from $\mathbb{R}$. More on that later.

Example 1.2 (Metric spaces). *Every metric space (X,d) gives a topology. Define τ_d by saying taht $U\subseteq X$ is open if for every $x\in U$ there exists $\varepsilon>0$ such that*

$$B(x;\varepsilon)\subseteq U.$$

Then τ_d is a topology on X. So every metric spaces is a topological space, but note that the opposite implication does not hold which is one of the reasons topology is more general.

Example 1.3 (The discrete topology). *Let X be any set. Let $\tau:=\mathcal{P}(X)$. Then τ is a topology on X which we call the* ***discrete topology****.*

Example 1.4 (The indiscrete topology). *Let X be any set. Define $\tau = \{\emptyset, X\}$. This is called the **indiscrete topology** or **trivial topology**.*

Definition 1.2 (Closed set). *A subset $C \subseteq X$ is called **closed** if its complement is open:*

$$C \text{ is closed} \Leftrightarrow X \setminus C \text{ is open.}$$

In a topological space, $\emptyset$ and X are closed. Arbitrary intersections of closed sets are closed and finite unions of closed sets are closed.

An important nuance is that closed does not mean not open. $\emptyset, X$ are both open and closed. In $\mathbb{R}$ (with the usual topology) we have that $\emptyset, \mathbb{R}$ are both open and closed while e.g. $[0, 1)$ is neither open nor closde.

Continuity

In Analysis, a function on metric spaces $f : X \to Y$ is continuous at $x \in X$ if small changes in x in $f(x) \in Y$. Recall the ε-δ definition:

$$\forall \varepsilon > 0, \ \exists \delta > 0 \text{ s.t. } d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Since we don't have access to metrics we must rewrite this.

Definition 1.3 (Continuity). *Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function*

$$f : X \to Y$$

*is **continuous** if for every open set $V \in \tau_Y$, the preimage*

$$f^{-1}(V)$$

is open in X.

Proposition 1.1. *Let (X, d_X) and (Y, d_Y) be metric spaces. A function*

$$f : X \to Y$$

is continuous in the ε-δ sense if and only if for every open set $V \subseteq Y$, the preimage

$$f^{-1}(V)$$

is open in X.

Proof. First suppose f is continuous in the ε-δ sense. Let $V \subseteq Y$ be open. We want to show that $f^{-1}(V)$ is open in X. Take $x \in f^{-1}(V)$.

Then $f(x) \in V$. Since V is open, there exists $\varepsilon > 0$ such that

$$B(f(x); \varepsilon) \subseteq V.$$

By continuity of f at x, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Thus, if $y \in B(x; \delta)$, then

$$f(y) \in B(f(x); \varepsilon) \subseteq V.$$

Therefore

$$y \in f^{-1}(V).$$

So

$$B(x; \delta) \subseteq f^{-1}(V).$$

Since every point of $f^{-1}(V)$ has a ball around it contained in $f^{-1}(V)$, the set $f^{-1}(V)$ is open.

Now suppose that for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X. Let $x \in X$, and let $\varepsilon > 0$. The ball

$$B(f(x); \delta)$$

is open in Y. Therefore

$$f^{-1}(B(f(x); \delta))$$

is open in Y. Since $x \in f^{-1}(B(f(x); \delta))$, there exists $\delta > 0$ such that

$$B(x; \delta) \subseteq f^{-1}(B(f(x); \delta)).$$

Thus, if

$$d_X(x, y) < \delta,$$

then

$$y \in f^{-1}(B(f(x); \delta)),$$

so

$$f(y) \in B(f(x); \delta).$$

Equivalently,

$$d_Y(f(x), f(y)) < \varepsilon.$$

Hence f is continuous at x. Since x was arbitrary, f is continuous. $\square$

Suppose τ_1 and τ_2 are two topologies on the same set X.

We say τ_1 is **finer** than τ_2 if

$$\tau_2 \subseteq \tau_1.$$

That means τ_1 has more open sets. Equivalently, τ_2 is **coarser** than τ_1. For every set X:

$$\{\emptyset, X\} \subseteq \tau \subseteq \mathcal{P}(X).$$

So the indiscrete topology is the coarsest topology possible, and the discrete topology is the finest topology possible.

This affects continuity. If the domain topology is finer, continuity is easier. If the target topology is finer, continuity is harder.

Example 1.5. *Let X have the discrete topology and let Y be any topological space. Then every $f : X \to Y$ is continuous.*
This is because $f^{-1}(V) \subseteq X$ and every subset of X is open.

Example 1.6. *Let Y have the indiscrete topology. Then every function $f : X \to Y$ is continuous.*
The preimages
$$f^{-1}(\emptyset) = \emptyset, \quad f^{-1}(Y) = x$$
are both open in X.

Example 1.7. *Let X be a set with two topologies τ_1, τ_2. Then* $\mathrm{id} : (X, \tau_1) \to (X, \tau_2)$ *is continuous exactly when $\tau_2 \subseteq \tau_1$.*

First glimpse of homeomorphisms

Topology is about spaces up to continuous deformation. The correct notion of "same space" is not equality, but **homeomorphism**.

Definition 1.4 (Homeomorphism). *Let X, Y be topological spaces. A function $X \to Y$ is a **homeomorphism** if*

1. *f is bijective,*
2. *f is continuous,*
3. *f^{-1} is continuous.*

If such an f exists, then X and Y are homeomorphic.

Intuitively, homeomorphic spaces have the same topological shape. Think about those videos of coffee cups and donuts. This is what they meant.

Example 1.8. *$(0,1)$ and $\mathbb{R}$ are homeomorphic. One possible homeomorphism is*
$$f : (0,1) \to \mathbb{R}, \quad x \overset{f}{\mapsto} \mathsf{tan}\left(\pi x - \frac{\pi}{2}\right).$$

So $\mathbb{R}$ is homeomorphic to $(0,1)$ even though $(0,1)$ is "small" (bounded) in comparison to $\mathbb{R}$.

Lastly we remark that openness doesn't need to have anything to do with distance. In algebraic geometry and commutative algebra we look at the prime spectrum of a ring,
$$\mathsf{Spec}(A)$$

which we endow with a topology called the **Zariski topology**. Closed sets are defined by ideals:

$$V(I) = \{\mathfrak{p} \in \mathtt{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

So the closed subsets are controlled by equations and ideals.

Exercises

Problem 1.1. *Let X be any set. Prove that the discrete topology $\tau = \mathcal{P}(X)$ is a topology on X.*

Proof. $\tau \subseteq \mathcal{P}(X)$ is ceirtainly satisfied. Furthermore $\emptyset, X \in \tau = \mathcal{P}(X)$ is also satisfied.

Now let I be some index set. Since $\mathcal{P}(X)$ contains all subsets of X and we necessarily have

$$\bigcup_{i \in I} U_i \subseteq X$$

for $U_i \in \mathcal{P}(X)$, and then we have

$$\bigcup_{i \in I} U_i \in \mathcal{P}(X) = \tau.$$

Thus τ is closed under arbitrary intersections.

Let $U_1, \dots, U_n \in \tau = \mathcal{P}(X)$. Once again since $U_i \subseteq X$ for all $1 \leq i \leq n$ we have

$$\bigcap_{i \in [n]} U_i \subseteq X$$

so

$$\bigcap_{i \in [n]} U_i \in \tau = \mathcal{P}(X).$$

□

Problem 1.2. *Let X be any set. Prove that $\tau = \{\emptyset, X\}$ is a topology on X.*

Proof. We have $\emptyset, X \in \tau$ by definition and also $\tau \subseteq \mathcal{P}(X)$.

Now notice that the only non-trivial collection of open sets in τ is the entireity of τ:

$$\emptyset, X,$$

and this collection is finite. Then $\emptyset \cup X = X \in \tau$ and $\emptyset \cap X = \emptyset \in \tau$.

Thus τ is closed under arbitrary unions and finite intersection adn hence forms a topology on X. □

Problem 1.3. *Let X be a topological space. Prove that arbitrary intersections of closed sets are closed.*

Proof. Let I be some index set so that C_i is closed for every $i \in I$. Then $X \setminus C_i$ is open for every $i \in I$. Then

$$\bigcup_{i\in I} X \setminus C_i = \bigcup_{i \in I} X \cap C_i^c$$

is open. Taking the complement we get

$$\left(\bigcup_{i\in I} X \cap C_i^c\right)^c = \bigcap_{i\in I} \emptyset \cup C_i = \bigcap_{i\in I} C_i$$

so

$$\bigcap_{i\in I} C_i$$

is closed. □

Problem 1.4. *Let τ_1, τ_2 be topologies on X. Prove that* $\mathrm{id} : (X, \tau_1) \to (X, \tau_2)$ *is continuous if and only if $\tau_2 \subseteq \tau_1$.*

Proof. Suppose $\tau_2 \subseteq \tau_1$. Let $U \in \tau_2$. Then $U \in \tau_1$ and

$$\mathrm{id}^{-1}(U) = U \in \tau_1.$$

Thus the preimage of $U \in \tau_2$ is in τ_1. As U was arbitrary we have that id is continuous.

Suppose id is continuous. Then for any open set $U \in \tau_2$ we have that

$$\mathrm{id}^{-1}(U) = U \in \tau_1$$

hence

$$\tau_2 \subseteq \tau_1.$$

□

1.1 Some exercises from Section 13 of Munkres

Problem 1.5 (Problem 1). *Let X be a topological space; let A be a subspace of X. Suppose that for each $x \in A$ there is an open set U containing x such that $U \subset A$. Show that A is open in X.*

Proof. For each $x \in A$, let U_x denote the open set containing x such that $U_x \subset A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

For each $x \in A$, $U_x \subset A$ so

$$\bigcup_{x \in A} U_x \subseteq A.$$

Now, for some arbitrary $x \in A$, there exists an open set U_x such that $x \in U_x$. Thus

$$A \subseteq \bigcup_{x \in A} U_x.$$

The two inclusions give the result. □

Problem 1.6 (Problem 3). *Show that the collection $\mathcal{T}_c$ given in Example 4 of §12 (of Munkres) is a topology on the set X.*
Is the collection

$$\mathcal{T}_\infty = \{U \mid X \setminus U \text{ is infinite or empty or all of } X\}$$

a topology on X?
Note: For a set X, define $\mathcal{T}_c$ to be the collection of all subsets U of X such that $X \setminus U$ is either countable or all of X.

Proof. Since $X \setminus \emptyset = X$ so $\emptyset \in \mathcal{T}_c$. Furthermore, since $X - X = \emptyset$ which is finite and thus countable, we have $X \in \mathcal{T}_c$.

Let $\{U_\alpha\}$ be a family of open sets (ignoring the trivial case where $U_\alpha = \emptyset$ for every such U_α) in $\mathcal{T}_c$. By definition, for each U, we have that

$$|X \setminus U_\alpha| \leq \aleph_0.$$

Since

$$\left|\bigcup_\alpha U_\alpha\right| \geq |U_\alpha|$$

for any α, we have

$$\left|X \setminus \bigcup_\alpha U_\alpha\right| \leq |X \setminus U_\alpha| \leq \aleph_0.$$

Hence $\bigcup_\alpha U_\alpha \in \mathcal{T}_c$ so $\mathcal{T}_c$ is closed under arbitrary unions.

Now let $\{U_i\}_{i \in [n]}$ be a finite family of open sets. Assume also that $U_i \neq \emptyset$ for every i as that case is trivial. Then

$$X \setminus \bigcap_{i=1}^n U_i = \bigcup_{i=1}^n (X \setminus U_i).$$

By assumption, $X \setminus U_i$ is countable and since the union of countable sets is countable we have that $\mathcal{T}_c$ is closed under finite intersections.

Now to answer the second question, $\mathcal{T}_\infty$ is absolutely not a topology on X. Take for instance $X = \mathbb{N}$. Then $\{n\} \in \mathcal{T}_\infty$, but

$$\mathbb{N} \setminus \bigcup_{n \in \mathbb{N}\setminus\{7\}} \{n\}$$

is $\{7\}$ which is neither infinite, empty, or all of X. □

Problem 1.7 (Problem 4).

a) If $\{\mathcal{T}_\alpha\}$ is a family of topologies on X, show that $\bigcup \mathcal{T}_\alpha$ is a topology on X. Is $\bigcup \mathcal{T}_\alpha$ a topology on X?

Proof of (a). Suppose $\{\mathcal{T}_\alpha\}$ is a family of topologies on X. Then $\emptyset, X \in \mathcal{T}_\alpha$ for all α so

$$\emptyset, X \in \bigcap \mathcal{T}_\alpha.$$

Let $\{U_\beta\}$ be a family of open sets in $\bigcap \mathcal{T}_\alpha$. Then every U_β lies in $\mathcal{T}_\alpha$ for every α. Thus

$$\bigcup U_\beta \in \mathcal{T}_\alpha, \ \forall \alpha$$

and hence $\bigcup U_\beta$ lies in the intersection.

Now let $\{U_i\}_{i \in [n]}$ be a finite family of open sets in $\bigcap \mathcal{T}_\alpha$.

Again, $U_i \in \mathcal{T}_\alpha$ for every i and α and the argument is analogous to the one above.

The same does not hold for $\bigcup \mathcal{T}_\alpha$. Suppose $X = \{1, 2, 3\}$ with $\mathcal{T}_1 = \{\emptyset, X, 1\}$ and $\mathcal{T}_2 = \{\emptyset, X, 2\}$. Then $\{1\}$ and $\{2\}$ lie in their union, but $\{1, 2\}$ does not. □

Problem 1.8 (Problem 6). *Show that the topologies $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are not comparable.*

Proof. Recall that the topology of $\mathbb{R}_\ell$ is

$$\mathcal{B}_\ell = \{[a, b) \mid a < b\},$$

and that the topology of $\mathbb{R}_K$ is

$$\mathcal{B}_K = \{(a, b) : a < b\} \cup \{(a, b) \setminus K : a < b\}$$

where

$$K = \left\{ \frac{1}{n} \,\middle|\, n \in \mathbb{Z}_+ \right\}.$$

These are not comparable. Take for instance $(-1,1) \setminus K \in \mathcal{B}_K$. If $\mathcal{B}_K \leq \mathcal{B}_\ell$ we should be able to find some $\varepsilon > 0$ so that $[0,\varepsilon) \subseteq (-1,1) \setminus K$, but for every $\varepsilon > 0$ we can choose n sufficiently large so that

$$\frac{1}{n} < \varepsilon.$$

Then $1/n \in [0,\varepsilon)$ implying $[0,\varepsilon) \not\subseteq (-1,1) \setminus K$. Thus $\mathcal{B}_K \not\subseteq \mathcal{B}_\ell$

For the other way consider $[0,1) \in \mathcal{B}_\ell$. Every K-neighbourhood at 0 is either (a,b) or $(a,b) \setminus K$ with $a < 0 < b$. Either way it contains some negative real numbers so no such basic neighborhood of 0 can fit in $[0,1)$. □

Problem 1.9 (Problem 8).

a) Apply Lemma 13.2 to show that the countable collection

$$\mathcal{B} = \{(a,b) \mid a < b,\ \ a,b \in \mathbb{Q}\}$$

is a basis that generates the standard topology on $\mathbb{R}$.

Proof of (a). Recall that Lemma 13.2 says that if X is a topological space, and $\mathcal{C}$ is a collection of open sets such that for each open set U of X and each $x \in U$ there is some element $C \in \mathcal{C}$ such that $x \in C \subset U$, then $\mathcal{C}$ is a basis for the topology on X.

Let U be open in the canonical topology on $\mathbb{R}$. Then $U = (a_0, b_0)$ for some $a_0, b_0 \in \mathbb{R}$ such that $a_0 < b_0$. By virtue of $\mathbb{Q}$ being dense in $\mathbb{R}$ we can, for each $\varepsilon > 0$ find some $a, b \in \mathbb{Q}$ suitably close:

$$a - a_0 < \varepsilon,\ \ b_0 - b < \varepsilon.$$

For any $x \in U = (a,b)$ let $\varepsilon = \min\{x - a_0, b_0 - x\} > 0$. Choose rational a and b (with $a < b$) so that $a - a_0, b_0 - b < \varepsilon$. Then

$$\begin{aligned} a - a_0 &< \varepsilon \\ &\leq x - a_0 \\ a &< x, \end{aligned}$$

and

$$\begin{aligned} b_0 - b &< \varepsilon \\ &\leq b_0 - x \\ -b &\leq -x \\ b &> x. \end{aligned}$$

Thus, by Lemma 13.2, $\mathcal{B}$ is a basis for the standard $\mathbb{R}$ topology.

This is a bit imprecise in places, but the idea is right. □